ARCHIVERSE PAKISTAN

90 s
0.3 m

The Station nightclub, Rhode Island, USA, 2003: smoke at 0.3 m in about 90 s. When was that decided?

Fire Safety from First Principles · Ep 00 · Swipe ->

#ArchiversePakistan

HOOK · THE STATION, RHODE ISLAND, USA, 20 FEB 2003

90 seconds from ignition to smoke at 0.3 m off the floor.

0 s	Pyrotechnics ignite polyurethane foam lining the platform walls and ceiling
41 s	Heat detection / alarm activates; crowd already moving to exits
~1 min	Smoke visible at the exit doorways
~90 s	Thick black smoke layer within about 0.3 m (1 ft) of the floor
<100 s	Main front doorway clogged with occupants
<5 min	Flames break through part of the roof

Full-scale mock-up by NIST, the US agency that investigated the fire

Unsprinklered: conditions at head height (1.5 m), 5.5 m from the fire, lethal in under 90 s.

NFPA 13 (US sprinkler standard) sprinklers: fire controlled, survivable at head height for the whole test (over 5 min).

Source: NIST NCSTAR 2, Vol. 1 (2005). 100 people died.

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THE CANVAS · SEASON 1, IN STRICT ORDER

26 episodes, from physics to a Pakistani checklist.

PHYSICS

- 00 Course map
- 01 Fire behaviour
- 02 Passive vs active

CONTAINMENT

- 03 Compartmentation
- 04 Fire resistance
- 05 Openings, firestopping
- 06 Fire, smoke curtains

MATERIALS

- 07 Reaction to fire
- 08 Fire-retardant
- 09 Façades

PEOPLE

- 10 Occupant load
- 11 Means of egress
- 12 Building types

SYSTEMS

- 13 Detection, alarm
- 14 Sprinklers, water
- 15 Special suppression
- 16 Firefighting access

PAKISTAN

- 17 The code
- 18 Design practice
- 19 Materials market
- 20 Façades in Pakistan
- 21 Firefighting in practice
- 21a Fire disasters, inquiries
- 21b Fire service capability
- 21c Private investment

CHECKLIST

- 22 Design checklist

Source: Archiverse programme backlog, S31 rows 00-22 (incl. 21a-21c). Order: our design.

BS 7974

Fire engineering starts with a design review.

British Standards from BSI, the UK's national standards body. BS 7974:2019: qualitative design review, quantitative analysis, assessment against criteria. BS 9999:2017: the whole life cycle.

Source: BSI Knowledge, BS 7974:2019 and BS 9999:2017 product pages



BEST PRACTICE · ENGLAND, UK

Three fire checks before anyone moves in.

Planning (2021) · Gateways 2 and 3 (Oct 2023)

Photo: Grenfell Tower, London, UK, 16 June 2017. ChiralJon, CC BY 2.0, via Wikimedia Commons

Source: gov.uk; Building Safety Regulator (England's new regulator for higher-risk buildings); Specification Online

FSP 2016

Pakistan's code already starts at the drawings.

Plans to the AHJ (authority having jurisdiction, e.g. building control) before construction (1.7.12.2). Construction documents must include all fire protection requirements (1.14.2). Fire certificate every 3 years (4.3.3).

Source: Building Code of Pakistan, Fire Safety Provisions 2016 (PEC)

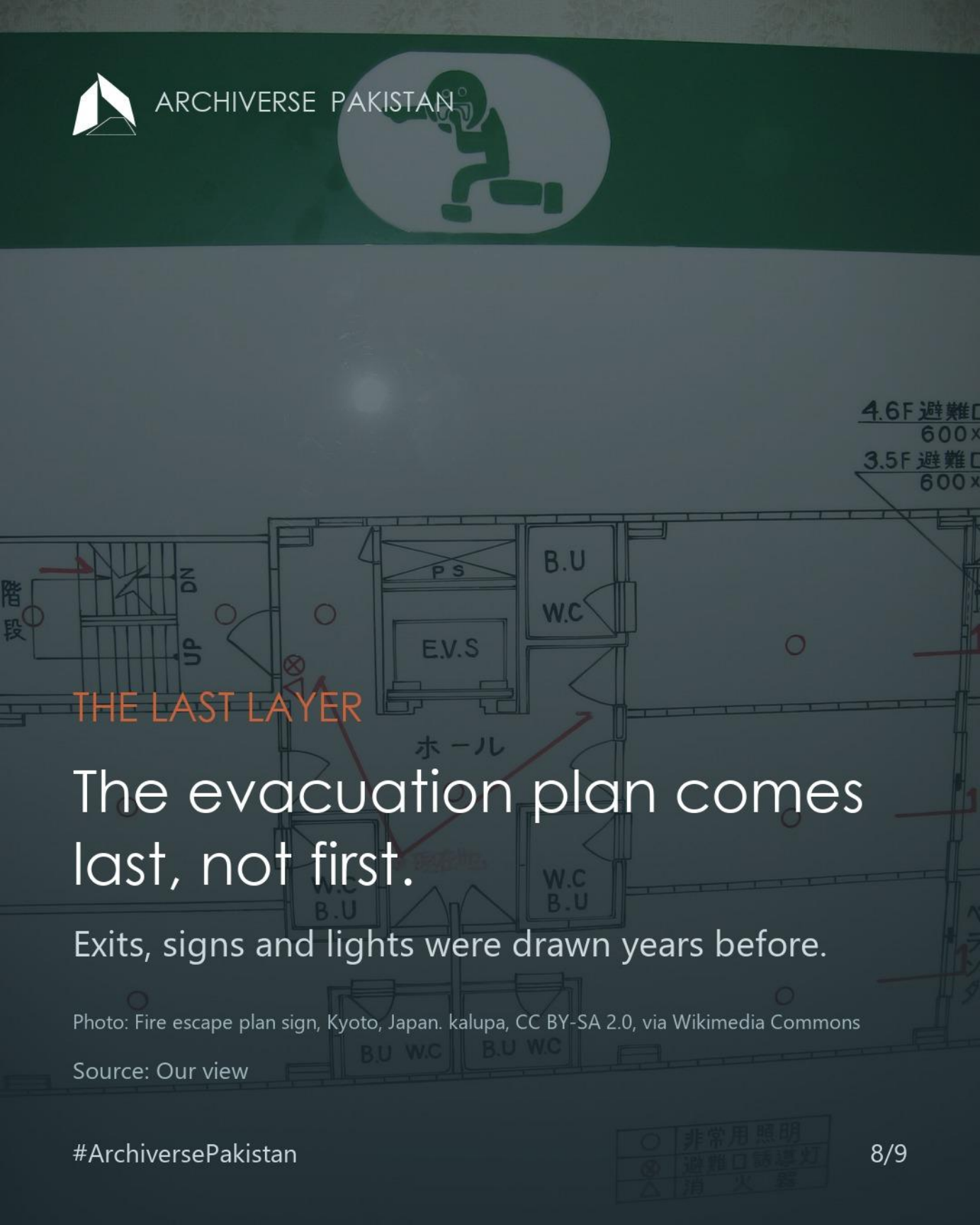
A PROPOSAL FOR DISCUSSION

Where fire safety is decided, and where the codes check it.

STAGE	WHAT IS FIXED (our reading)	ENGLAND, UK	PAKISTAN FSP
Brief and concept	Occupancy, occupant load, compartments, escape routes: the fire strategy		
Planning	Height, massing, access for firefighting	Gateway 1: fire statement (from 1 Aug 2021)	
Detailed design	Fire resistance, finishes, systems, smoke control		1.14.2: all fire protection in documents
Approval before work	Checked drawings and calculations	Gateway 2: BSR approval (from 1 Oct 2023)	1.7.12.2: plans to AHJ before construction
Construction	Built as drawn: doors, firestopping, linings		1.14.2(3): approved set on site
Completion	Information to the operator	Gateway 3 certificate; Reg. 38 handover info	
Use	Management, drills, the evacuation plan (last)		4.3.3: fire certificate every 3 years

Proposal: a one-page fire strategy drawing in every FSP 1.14 submission.

Source: gov.uk; BSR; Building Regulations 2010 (England) Reg. 38, Approved Document B; PEC FSP 2016



THE LAST LAYER

The evacuation plan comes last, not first.

Exits, signs and lights were drawn years before.

Photo: Fire escape plan sign, Kyoto, Japan. kalupa, CC BY-SA 2.0, via Wikimedia Commons

Source: Our view

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At which stage was your last fire strategy drawn?

And who drew it? Tell us in the comments.
Next: Ep 01, fire behaviour.
#ArchiverseAccountable

Full piece
Link in bio

Learn. Question. Build.